Data Modelling & HLD (High level Design)

(DM)

System Design

LLD + HLD + DM

- Intro.
- SQL
  - Keys
  - Normaliz^{n}
  - ACID
- Entity / Entity Rel^{n} / Attr.
- SQL Internals
  - Indexe
  - How to write efficient queries.
  - Query optimiz^{n}.

Data Modelling → organizing data
- store
- fetch efficiently

Any piece of information.

Data is king

Data is new oil.

What kind of data do we use on a daily basis?

Q What kind of data do we use on a daily basis?

- images
- to-do lists
- expenses
- contacts
- ⋮

Q Appl^{n} s/w to store data

- notepad
- google sheets
- excels
- ⋮

AirTribe

students (excel-sheets)

id	name	psp	attendance	college	org.	Batch id
1	Rahul	80	150	1		B17
2	Simran	70	93	1		C19
3	Sahil	100	7	1	1	C19
4	Rahul	29	68	1	1	C18
5	Abhinav	84	71			C20
⋮	⋮					⋮

↳ Find avg. attendance of students ? → easily achievable from excel in-built fn

↳ Find avg. attendance corresponding to each batch.

More & more req. become complex

↳ excel sheet won't be able to handle those.

Concl:

Although we have variety of s/w to store data in required manner, not many can support complex extractions.

⊛ Drawbacks

① Inefficient

10k records
↓
1M
↓
10M

extraction time ↑ exponentially

② Data Integrity

id	name	psp	attendance	Batch_id
1	Rahul	80	~~150~~ Full	A01 → doesn't exist
2	Simran	70	93	
3	Sahil	~~100~~ Topper	~~7~~	
4	Rahul	29	68	
5	Abhinav	84	71	
⋮	⋮			

↳ Inaccurate / Corrupted

Data is still meaningful.

BUT we can't perform operations (Eg. avg attendance)

3. Concurrency

↓

If multiple ppl work on the same thing (data) together

↓

Inconsistency.

4. Security Issues

↳ Level of Access (granularity)

↳ Encryption

||

Sol^{n} :— Database.

(K) What is Database ??

↳ collection of related data.

{ Airbase, Army base, Naval base } → everything related to something interrelated

Naval base

somehow interrelated

e.g. AirTribe → students, batch, instructor

⋮

✳ DBMS (Database Management Sys.)

A s/w sys that allows you to efficiently manage a DB

↓

no drawbacks as discussed above

✳ Operations to perform on a DB (using DBMS)

↳ CRUD

- ↳ Delete
- Update
- Read / Fetch
- Create

✳ Type of DB's

⊛ Type of DB's

2 Types

- Relational
- Non-relational

⊛ Relational

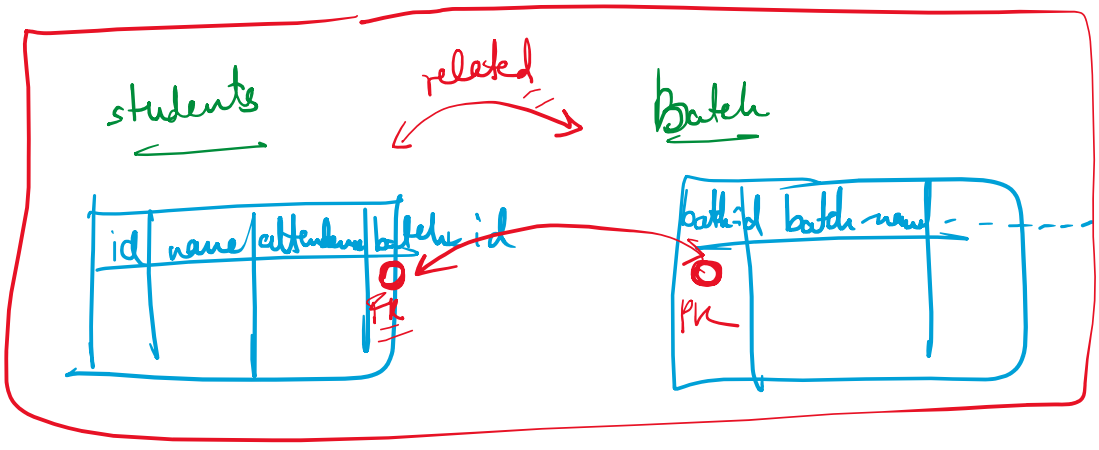
All relational DB's are structured but vice versa not necessarily true :)

allows you to store data in form of relations

↓

tables + cols.

[(rows) + (cols)]



All relational DB follow ACID prop.

All relational DB follow ACID prop.
but vice-versa not necessarily true

(*) Prop. of RDBMS → Eg. SQL

1. It represents data in form of tables
(collection of tables)
w/ each table (entity) storing info about something
- entity
- relationships

2. Every row is unique

name	psp	attendance
→ Raghu	73	100
→ Shyam	81	83
→ Veeru	100	42
→ Salib	23	29
→ Ashish	98	76
→ Raghu	73	100

appear to be redundant

∴ No two rows should have exactly same value in all cols.

why ?

Why ?

~~Find~~ psp of Rahul ?

↳ which Rahul

↓

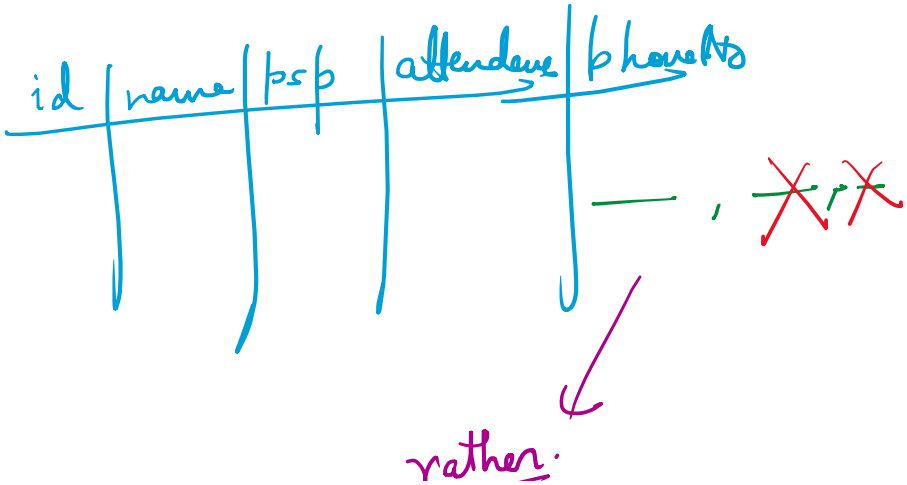
Note !- This is the reason by default generally (==) id

every row has a unique value (PK) across that col. in the table. =

(3) A col. should have all values of same data-type

↳ helps to maintain data-integrity

(4) All values/cell should be atomic



rather.

style CSV {—, —, —}

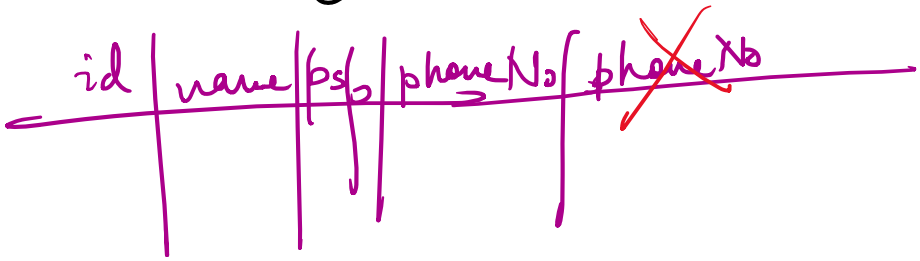
(5) Seq. of col. is not guaranteed in RDBMS

Note!- MySQL preserves order as of create command.

(6) Seq. of rows is not guaranteed by RDBMS.

Note: MySQL preserves

(7) Name of any col. is unique



Q Why?

↳ Ambiguity

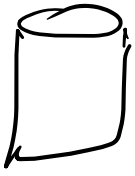
(A) Scaling → to increase.

↓

In size

↙ ↘

Horizontal Vertical



→ Amount of Data ∝ size of Disk

DB

256 GB → 1 TB

↙ ↘

Inc. the storage cap. of my m/c ie, 2 TB

↓

Vertical scaling

(In context of DB)

(B) Store the extra Δ (delta) in another m/c.

↓

Horizontal scaling

↓

DB)

↓ inc. amt. of res. in same m/c

↓ add more m/c w/ same resources

So as to

inc → computational cap.

→ storage cap

⋮

Myth

~~Rel^nal DB are vertically scalable~~

~~Non-rel^al DB " horizontally "~~

Eg → Sharding

⊛ When to use Relational DB ?

are too

Relational DB's :

- Read patterns are too complex (lot of JOINs)
- Strong consistency
- ACID prop.
- Fixed st. / Schema.